SIMATS ENGINEERING

Engineer to Excel

SIMATS

Saveetha Institute of Medical And Technical Sciences

(Declared as Deemed to be University under Section 3 of UGC Act 1956)

CSA0408 - OPERATING SYSTEMS

CO1 – AT3

TECHNICAL PRESENTATION

Question 3 scenario: Real-time embedded system used in traffic signal control.

Presented by:

Subash G

Reg No - 192572150

Guided by:

Dr. Vasavi J

Real-Time Traffic Signal Control

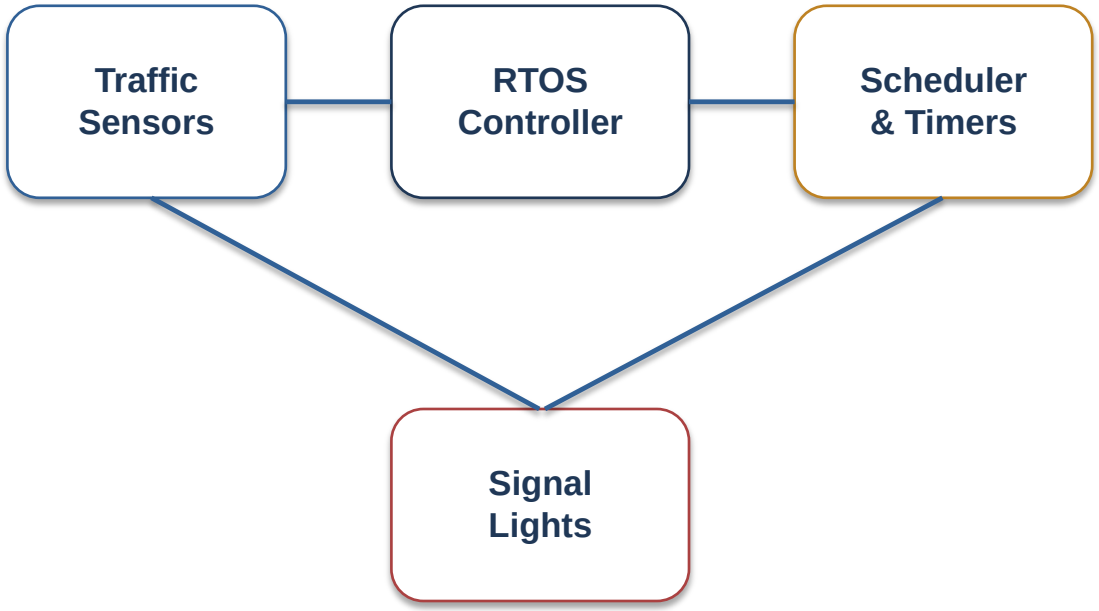
Problem Scenario

A traffic signal controller must respond to vehicle detection, timing events and emergency conditions within predictable time limits.

Key requirement	**Predictable response + correct signal timing**



System flow



Sensors provide events → RTOS schedules tasks → controller updates signals

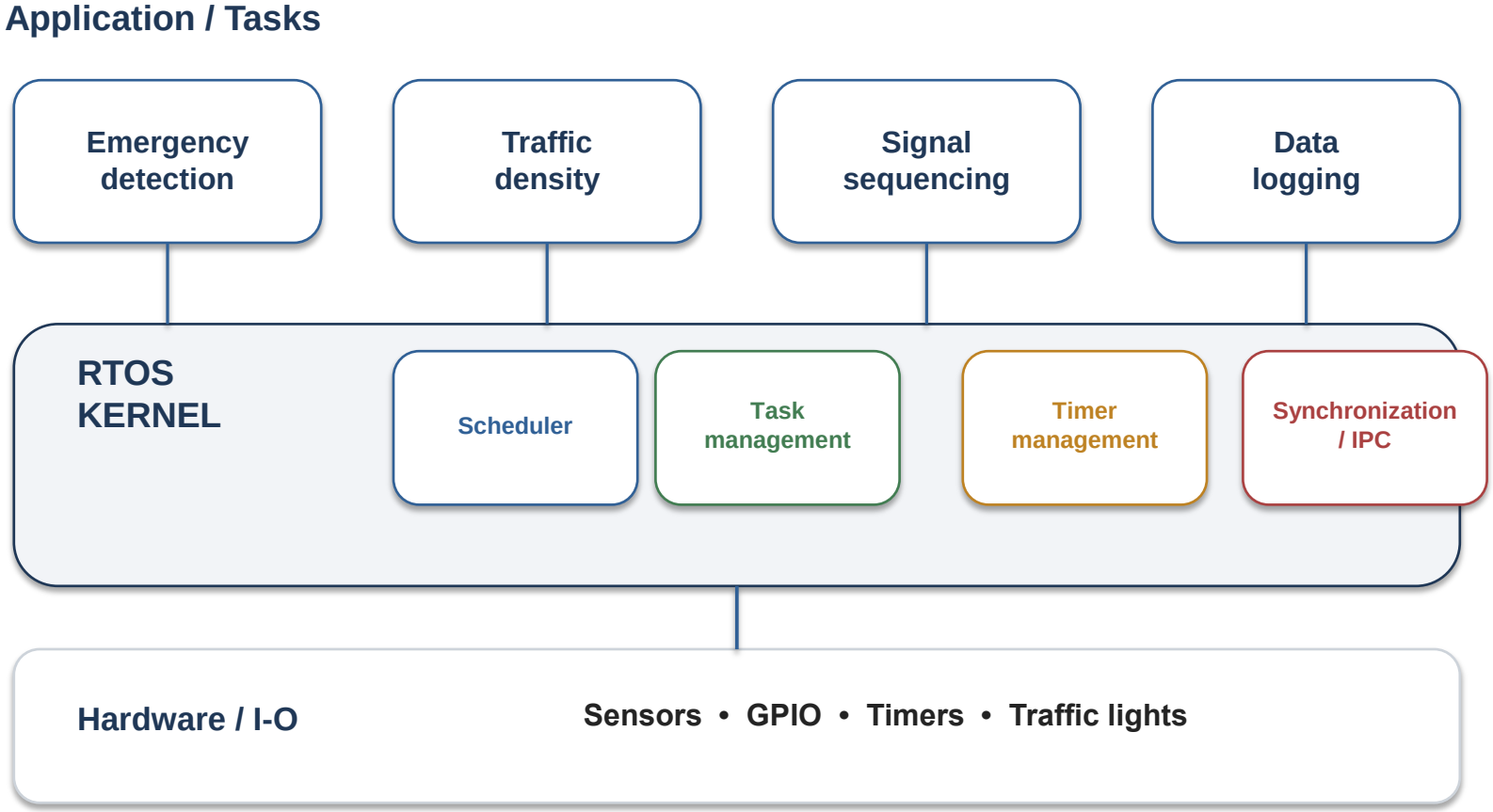
Why a Real-Time Operating System?

Traffic signal control is deadline-sensitive: the system must react within a predictable time.

Requirement	General-purpose OS	RTOS	Traffic signal need
Response time	Can vary	Predictable	Fast, bounded response
Scheduling	General purpose	Priority-based	Urgent events first
Deadlines	Not always guaranteed	Designed for deadlines	Timing must be respected
Typical use	PCs / general servers	Embedded control	Traffic controller

Conclusion: RTOS is appropriate because predictable timing and priority-based response are central to the application.

RTOS Architecture for Traffic Control



How it works

1. Sensors create events
2. Tasks request kernel services
3. Scheduler selects the next task
4. Timers maintain signal phases
5. I/O updates the lights

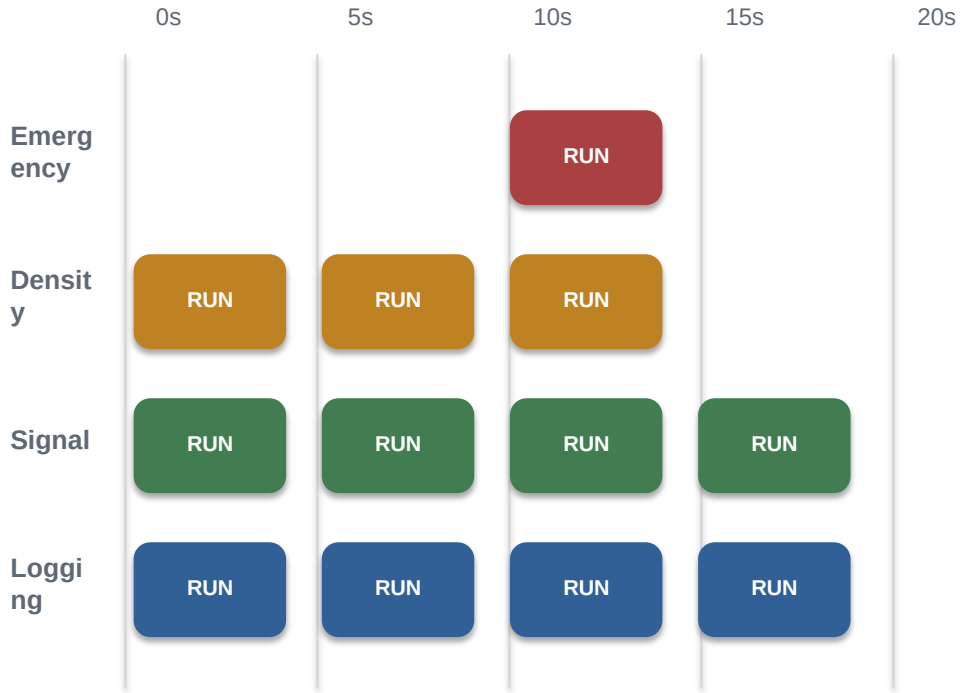
Task Scheduling and Priority

Process/task control in the traffic controller

Higher-priority events can require immediate attention.

P1	**Emergency vehicle detection**	Immediate signal override
P2	**Traffic density monitoring**	Adjust phase duration
P3	**Signal sequencing**	Normal light cycle
P4	**Traffic data logging**	Background task

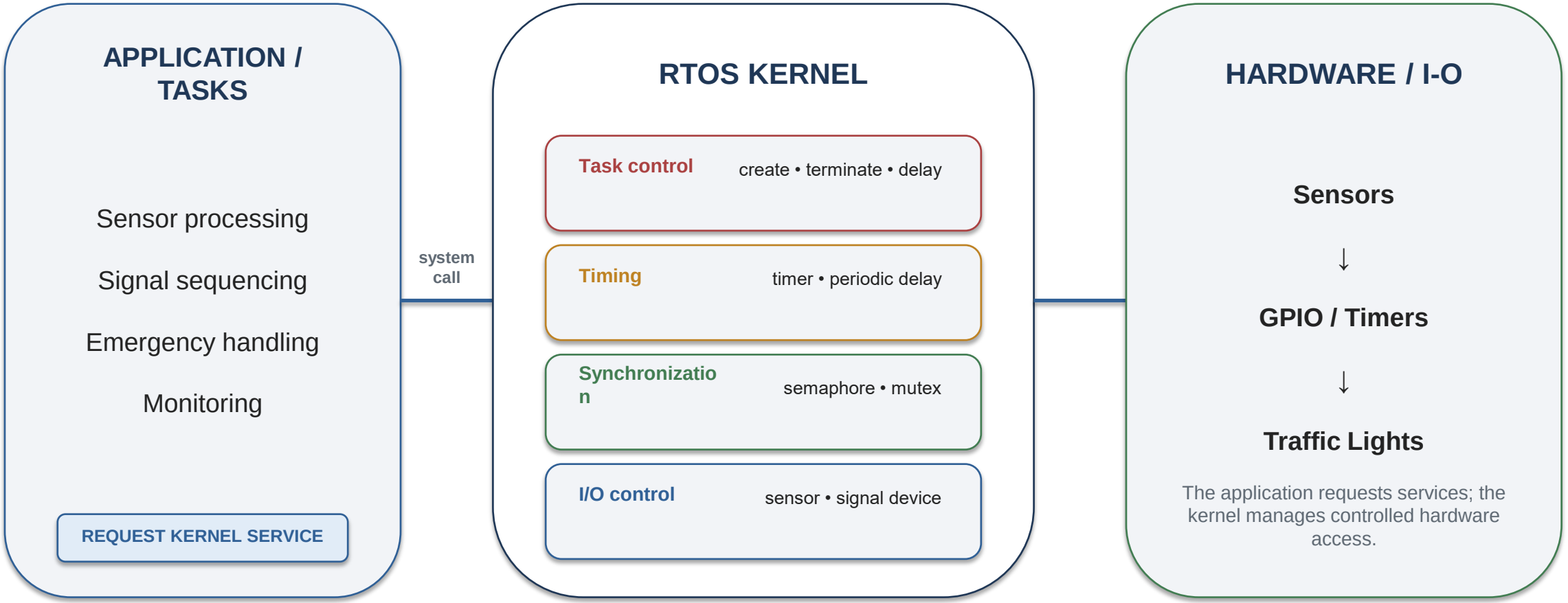
Illustrative execution timeline



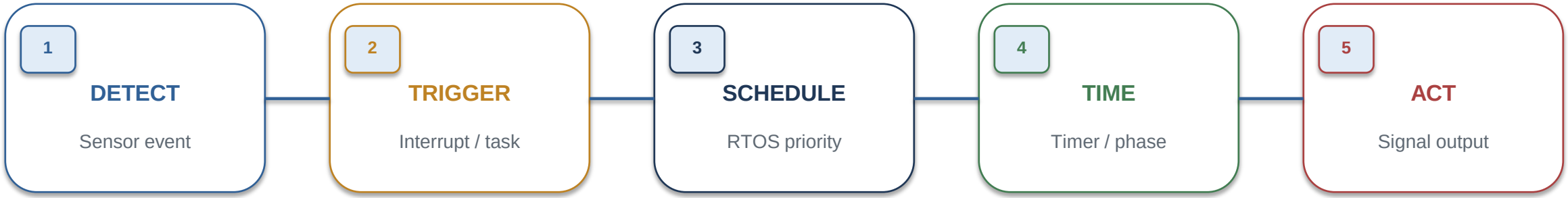
Example: emergency task pre-empts normal work.

System Calls and Timing

Application tasks interact with the kernel through controlled services



Complete Working Model



What the system demonstrates

OS type	Real-Time Operating System
Core mechanism	Priority scheduling + timers
Kernel interaction	System calls for control and timing

FINAL TAKEAWAY

RTOS combines predictable scheduling, timing and kernel services to control traffic signals.

A practical OS design for a deadline-sensitive embedded system.


SIMATS ENGINEERING
SIMATS
CSA0408 - OPERATING SYSTEMS
CO1 – AT3
TECHNICAL PRESENTATION
Question 3 scenario: Real-time embedded system used in traffic signal control
Presented by:
Subash G
Reg No - 192572150
Guided by:
Dr. Vasavi J
408
GPS Map Camera
Google
Mevalurkuppam, Tamil Nadu, India
22g8+cwh, Mevalurkuppam, Tamil Nadu 602105, India
Lat 13.02652° Long 80.01714°
Saturday, 03/10/2026 01:51 PM GMT +05:30